Zhaoxiang Feng

Open to relocation | (336) 337-7139 | aaronzhfeng@utexas.edu | GitHub: github.com/aaronzhfeng
Available: May–August 2027 (12+ weeks, full-time)

EDUCATION

The University of Texas at Austin — Austin, TX
Master of Science in Computer Science — *08/2026–05/2028*

University of California San Diego — San Diego, CA
B.S. in Data Science; B.S. in Probability & Statistics (GPA: 3.84/4.00) — *09/2022–03/2026*
Related Coursework: Machine Learning, Deep Learning, Data Structures and Algorithms, Data Management, Time Series Analysis

RESEARCH EXPERIENCE

Hao AI Lab, UC San Diego — San Diego, CA
Research Assistant (Advisor: Prof. Hao Zhang) — *09/2025–Present*

- Built the standalone **LLM inference engine** for **JetSpec** (parallel tree speculative decoding): **paged KV cache**, a continuous-batching scheduler, a paged **tree-attention Triton kernel**, and **CUDA-graph** capture for verification, sustaining **∼1,000 tok/s** (peak 1,456) on a single **NVIDIA B200** with Qwen3-8B.
- JetSpec reaches up to **9.64× end-to-end speedup** over autoregressive decoding on Qwen3-8B (MATH-500); second author, under review at **NeurIPS 2026**, 177 GitHub stars.
- Led **JetSpec++**, a hidden-space **attention-layer causal correction** for parallel drafting; at matched size (75.5M vs. 78M parameters) it accepts **0.89 more tokens per verification round** than DeepSeek's DSpark, and the lead grows to **1.67** at unseen draft depths without retraining.
- Ported **DFlash** block-diffusion speculative decoding to **Google Cloud TPU** in **JAX** as primary contributor: **3.13× average speedup** across 9 benchmarks on TPU v5p (up to 5.72×, TPOT 8.02→1.40 ms) and **2.29× vs. 1.30×** for EAGLE-3; featured on the **Google Developers Blog**.
- Upstreamed the port to **vLLM tpu-inference** in 3 PRs (model and proposer merged; pipeline integration and CI in review), with JAX unit tests and a reproducible benchmark harness.

Q-Lab, UC San Diego — San Diego, CA
Research Assistant (Advisor: Prof. Lianhui Qin) — *01/2025–06/2026*

- Contributed to **ArcMemo**, concept-level lifelong memory for LLM reasoning; **Runner-Up, ARC Prize 2025 Paper Awards** (top 8 of 90).
- Built a **concept-to-puzzle generation pipeline** in which an LLM turns each hand-written concept into executable puzzle-generating code, filtered for determinism, well-formedness, and color invariance.
- Worked on **reasoning-guided retrieval** replacing embedding search, and designed the method's adaptation to **AIME** competition math with memory-format ablations.

Wang Lab, UC San Diego — San Diego, CA
Student Researcher (Advisor: Prof. Wei Wang) — *06/2025–06/2026*

- Developing **TrustPPI**, a model-agnostic reliability signal for protein interaction models based on **embedding-perturbation stability**; reaches **0.66–0.83 AUROC** for error detection across 3 architectures, where native confidence scores sit near 0.5.

PROJECTS

LLM Pre-training System | Senior Capstone, UC San Diego (Advisor: Prof. Hao Zhang) — *09/2025–12/2025*

- Built a training system for a **1.8B-parameter** Qwen3-style model on **8× NVIDIA B200** with **DDP + ZeRO-1**, **FlashAttention-2**, **BF16**, **torch.compile**, and FLOPs-accurate MFU monitoring.
- Trained on **117B tokens** within a 60-hour budget at **160K tokens/s** and **36.7% MFU**; implemented **SFT** and **DPO** post-training, improving benchmark accuracy by up to 2.6%.

TECHNICAL SKILLS

Programming Languages: Python, Triton, R, SQL, Java, JavaScript
ML Systems & Inference: vLLM, speculative decoding, paged KV cache, continuous batching, CUDA graphs, FlashAttention-2
Accelerators & Distributed: NVIDIA B200/H100, Google Cloud TPU (v5p, v6e), JAX, PyTorch DDP, ZeRO, BF16 mixed precision
ML & AI: PyTorch, Hugging Face Transformers, SFT, DPO, LLM agents and memory, RAG
Tools: Git, Linux, Docker, Hydra